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Basics of Backprojection Algorithm for Processing Synthetic Aperture Radar Images

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Abstract

The purpose of this report is to provide a background to Synthetic Aperture Radar (SAR) image formation using the Filtered Backprojection (FBP) processing algorithm.

This is meant to be an aid to those tasked to implement real-time image formation using the Filtered Backprojection processing algorithm.

Acknowledgements

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Foreword

This report details the results of an academic study. It does not presently exemplify any modes, methodologies, or techniques employed by any operational system known to the author.

Classification

The specific mathematics and algorithms presented herein do not bear any release restrictions or distribution limitations.

This distribution limitations of this report are in accordance with the classification guidance detailed in the memorandum “Classification Guidance Recommendations for Sandia Radar Testbed Research and Development”, DRAFT memorandum from Brett Remund (Deputy Director, RF Remote Sensing Systems, Electronic Systems Center) to Randy Bell (US Department of Energy, NA-22), February 23, 2004. Sandia has adopted this guidance where otherwise none has been given.

This report formalizes preexisting informal notes and other documentation on the subject matter herein.

1 Introduction

Synthetic Aperture Radar (SAR) image formation is essentially the implementation of a matched filter to collected radar echo data for an array of target locations, where each target location is a pixel in a SAR image. Most common image formation algorithms approximate the matched filter to some degree in order to achieve some degree of processing efficiency for real-time implementations. This applies to algorithms like the Polar Format Algorithm (PFA), and Overlapped Subaperture Algorithm (OSA), and all perturbations and extensions of these. The advent of improved processors allows employing image formation algorithms that are less efficient but offer higher fidelity, more accurately implementing matched filter performance over a larger image area and at finer resolutions, even for real-time applications. This is the case for a class of algorithms known as Backprojection (BP), also known as Filtered Backprojection (FBP), or Convolution Backprojection (CBP).

There is nothing particularly special about FBP processing. Although as a real-time SAR image formation algorithm it has only relatively recently become viable, FBP is inherently able to effectively deal with range migration, particularly a problem for fine-resolution processing. Particularly important, FBP is well understood and well documented. Lack of suitable documentation is a shortcoming of some of its rivals. Nevertheless, this report does not presume to imply that FBP is the ‘best’ choice for any particular SAR mode implementation; only that it will work, and below is presented how to do that.

As background for this report, we offer the following:

BP algorithms have their roots in Computer Tomography, which is discussed in some detail in an earlier report by Doerry.¹

The earliest publication that posits BP as a viable SAR image formation algorithm was by Munson, et al.²

An excellent primer on implementing BP using Matlab is given by Gorham and Moore.³

We opine that valuable insight can often be gained by analyzing other SAR image formation algorithms. Of particular note is PFA processing, originally developed by Walker,⁴ and also discussed in detail in a Sandia National Laboratories report by Doerry.⁵ This algorithm is very popular and treated in a number of good texts.^{6,7}

Many other publications are readily available about various BP implementations and related topics. We do not wish to shortchange or slight these articles, but merely indicate that there are too many to list. Specific other relevant reports will also be identified in later sections.

*“Any fool can know. The point is to understand.”
— Albert Einstein*

2 The Data Model

We review and abridge elements of earlier reports.

With spotlight mode SAR, including individual spotlight images that become mosaicked into stripmaps, typically each pulse is identified with a unique pulse number or index, and used in a single SAR image. The pulse index number is known prior to collection. This mode allows the maximum degree of motion compensation via pulse parameter modulation, if so desired.

With VideoSAR modes, also sometimes referred to as Circular SAR, or Motion Imagery SAR (MISAR), an individual pulse may be used in more than one SAR image, and may have different index values assigned after collection depending on what subset of pulses are selected for a specific SAR image patch. This mode limits the amount of effective motion compensation via pulse parameter modulation. For many image formation algorithms, this requires more phase error compensation than legacy spotlight modes. For FBP algorithms, it is largely inconsequential to the efficiency of the algorithm.

2.1 The Radar Echo Signal Model

Everything begins with a signal model. We shall assume a transmitted (TX) signal of the form

$$x_T(t, n) = A_T \operatorname{rect}\left(\frac{t - t_n}{T_{TX}}\right) \cos(\Phi_T(t - t_n, n)), \quad (1)$$

where

$$\begin{aligned} n &= \text{azimuth sample index with bounds, } -N/2 \leq n < N/2, \\ N &= \text{total number of azimuth samples, presumed even for convenience,} \\ t_n &= \text{reference time for the } n\text{th TX pulse,} \\ A_T &= \text{arbitrary amplitude of the TX pulse,} \\ T_{TX} &= \text{the TX pulse width, and} \\ \Phi_T(t, n) &= \text{phase function of the TX pulse,} \end{aligned} \quad (2)$$

and the pulse envelope function

$$\operatorname{rect}(z) = \begin{cases} 1 & |z| \leq 1/2 \\ 0 & \text{else} \end{cases}. \quad (3)$$

For a Linear Frequency Modulated (LFM) chirp, the phase function is identified as

$$\Phi_T(t, n) = \phi_{T,n} + \omega_{T,n}t + \frac{\gamma_{T,n}}{2}t^2, \quad (4)$$

where

$$\begin{aligned} \phi_{T,n} &= \text{reference phase for } n\text{th pulse,} \\ \omega_{T,n} &= \text{reference center frequency for the } n\text{th pulse, and} \\ \gamma_{T,n} &= \text{reference chirp rate for the } n\text{th pulse.} \end{aligned} \quad (5)$$

Other phase functions can, and often are, used, but we shall herein continue with the LFM chirp.

The received echo signal from a single point target reflector is then

$$x_R(t, n) = A_R \text{rect}\left(\frac{t - t_n - t_{s,n}}{T_{TX}}\right) \cos\left(\Phi_T(t - t_n - t_{s,n}, n)\right), \quad (6)$$

where

$$\begin{aligned} A_R &= \text{the received echo signal amplitude, and} \\ t_{s,n} &= \text{the echo delay time for the point target scatterer.} \end{aligned} \quad (7)$$

For convenience and clarity, we will hereafter assume the “stop & go” model for motion, where the echo delay time is assumed to be constant during the received echo pulse. This is often sufficient for practical implementations, but if necessary, modifications to treat intra-pulse Doppler are fairly minor.

The received signal is typically down-converted to baseband with a Local Oscillator (LO) signal modeled as

$$x_L(t, n) = 2 \text{rect}\left(\frac{t - t_n - t_m}{T_{LO}}\right) \exp(j \Phi_L(t - t_n - t_m, n)), \quad (8)$$

where

$$\begin{aligned} t_m &= \text{reference time offset for the } n\text{th LO pulse,} \\ T_{LO} &= \text{the LO pulse width, and} \\ \Phi_L(t, n) &= \text{phase function of the LO pulse.} \end{aligned} \quad (9)$$

Quadrature demodulation then yields a baseband video signal of the form

$$x_V(t, n) = A_R \text{rect}\left(\frac{t - t_n - t_{s,n}}{T_{TX}}\right) \text{rect}\left(\frac{t - t_n - t_m}{T_{LO}}\right) \exp\left(j \begin{pmatrix} \Phi_T(t - t_n - t_{s,n}, n) \\ -\Phi_L(t - t_n - t_m, n) \end{pmatrix}\right), \quad (10)$$

where the subscript “V” here means “video” signal. This is just a product of the amplitudes and a difference in the phases.

More generally, the radar echo signal is a superposition of distributed and multiple discrete scatterers, and is more properly written as

$$x_V(t, n) = \int_{\mathbf{s}} \left\{ \sigma(\mathbf{s}) \text{rect}\left(\frac{t - t_n - t_{s,n}}{T_{TX}}\right) \text{rect}\left(\frac{t - t_n - t_m}{T_{LO}}\right) \times \exp\left(j \begin{pmatrix} \Phi_T(t - t_n - t_{s,n}, n) \\ -\Phi_L(t - t_n - t_m, n) \end{pmatrix}\right) \right\} d\mathbf{s}, \quad (11)$$

where

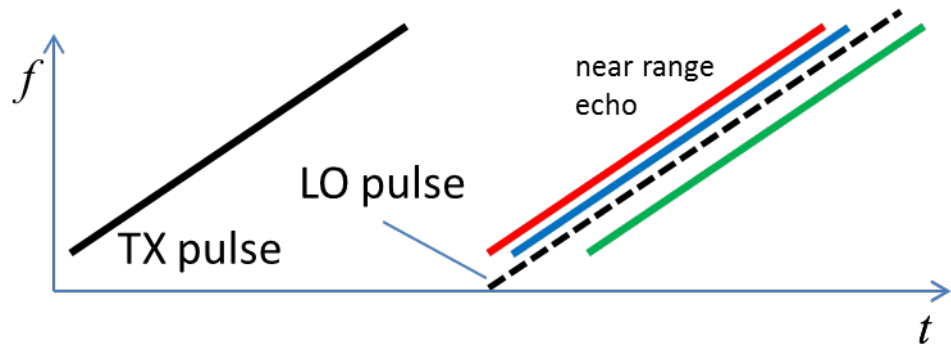
$$\begin{aligned} \mathbf{s} &= \text{a vector defining a target location in the field of view of the radar, and} \\ \sigma(\mathbf{s}) &= \text{amplitude function of the target echoes in the scene.} \end{aligned} \quad (12)$$

The amplitude function $\sigma(\mathbf{s})$ embodies not only the intrinsic reflectivity of the target scene, but any antenna beam illumination functions, and power/energy losses with respect to range as well.

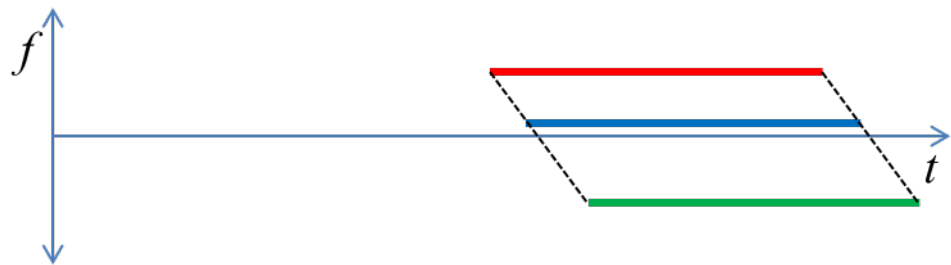
Some comments on the LO phase function are in order.

- If $\Phi_L(t, n)$ exhibits a constant frequency, then the baseband video signal is merely a down-shifted copy of the received signal’s spectrum. This is generally the case when modulations other than LFM chirp are used for the TX signal, but can also be the case when an LFM chirp is used for the TX signal. This is often the preferred method for wide range-swaths and coarse range resolutions.
- When the TX signal is an LFM chirp, another option for the LO signal is often employed, where the LO signal is itself a LFM chirp. This is called “stretch” processing.⁸ Effectively, the video signal is “de-chirped.” This can often reduce the video signal bandwidth, allowing easier digital sampling of the signal. This is often the preferred method for narrow-swaths with fine range resolution. Figure 1 illustrates the modulations employed for stretch processing.

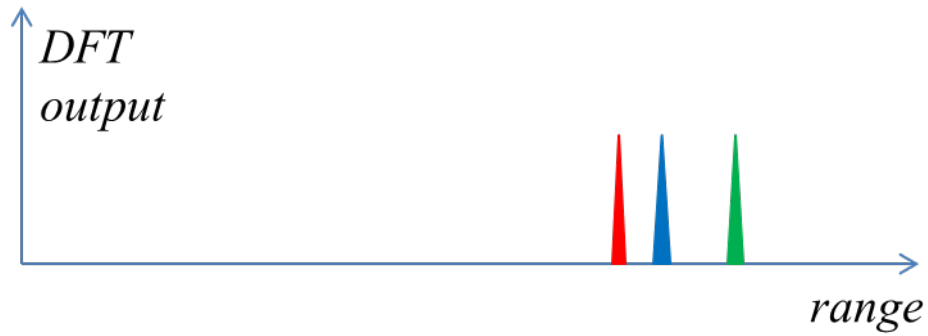
Effective FBP processing can be applied in either case, and with any waveform suitable for SAR. The FBP processing differences will be in the preprocessing stage of the overall algorithm. We shall proceed assuming stretch processing with an LFM chirp waveform.



(a) Microwave Signals



(b) Baseband Video Signals



(c) Range-Compressed Signals

Figure 1. Stretch Processing. After de-ramping the echo energy with a LO pulse that is also a chirp, range is encoded in baseband video frequency. Consequently a Discrete Fourier Transform (DFT) will separate frequencies, and hence ranges.

Furthermore, we will assume that the LO signal has the same reference phase, reference frequency, and constant chirp rate as the TX signal. Furthermore yet, we will assume the reference time t_m for the LO signal is adjusted to be the echo time delay to a single reference location in the target scene, which we term as the Scene Reference Point (SRP). That is

$$t_m = t_{c,n} = \text{echo delay time to SRP for the } n\text{th pulse.} \quad (13)$$

Note that the reference time therefore changes as a function of the pulse index. The resulting video signal is then sampled offset from reference time t_n , specifically at times

$$(t - t_n - t_{c,n}) = T_s i, \quad (14)$$

where

$$\begin{aligned} i &= \text{intra-pulse, fast-time sample index, } -I/2 \leq i < I/2, \\ I &= \text{total number of fast-time samples, presumed to be even for convenience, and} \\ T_s &= \text{nominal fast-time sample spacing in sec.} \end{aligned} \quad (15)$$

We will make the common assumption hereafter that all samples are taken during the interval defined by the overlapped returns of all targets at all ranges of interest. In other words, a target at any location of interest will return energy in all data samples. No data samples are “empty” for any potential target of interest.

We return to the model of a single point target scatterer of Eq. (10), and refine it with our new assumptions to yield

$$x_V(i, n) = A_R \exp \left\{ j \left(-(\omega_{T,n} + \gamma_{T,n} T_s i)(t_{s,n} - t_{c,n}) + \frac{\gamma_{T,n}}{2} (t_{s,n} - t_{c,n})^2 \right) \right\}. \quad (16)$$

Note that all information with respect to the location of the target scatterer is in the phase.

It is often useful, particularly for real-time motion compensation,⁹ for a radar to modulate its waveform parameters as a function of collection geometry in a manner such that

$$\begin{aligned} \omega_{T,n} &= \omega_0 \kappa_n, \text{ and} \\ \gamma_{T,n} &= \gamma_0 \kappa_n, \end{aligned} \quad (17)$$

where

$$\begin{aligned} \omega_0 &= \text{nominal constant reference frequency for the synthetic aperture,} \\ \gamma_0 &= \text{nominal constant reference chirp rate for the synthetic aperture, and} \\ \kappa &= \text{'wavenumber projection compensation factor'}. \end{aligned} \quad (18)$$

We will discuss the parameter κ in greater detail later.

We will also find it useful to define the relative delay term

$$\tau_{sc,n} = (t_{s,n} - t_{c,n}) = \text{relative delay between scatterer and SRP.} \quad (19)$$

This allows us to arrive at a model for the phase of an echo signal from a point scatterer in the demodulated base-band data as

$$\Phi_V(i, n) = \left(-(\omega_0 + \gamma_0 T_s i) \kappa_n \tau_{sc,n} + \frac{\gamma_0 \kappa_n}{2} (\tau_{sc,n})^2 \right). \quad (20)$$

The overall video echo signal record is termed the Phase History (PH) data. Consequently, Eq. (16) is the general PH data model, but if we use parameter modulation given by Eq. (17), the phase can be written in a convenient form given by Eq. (20).

2.2 Geometry

With a specific set of pulses defining the synthetic aperture selected, we can define a particular coordinate frame for that synthetic aperture as indicated in Figure 2.

We now define a somewhat arbitrary x, y, z coordinate frame. The xy plane is locally horizontal to the Scene Reference Point (SRP). The SRP defines the center or origin of the x, y, z coordinate frame. It is also sometimes referred to as the Motion Compensation Point (MCP) because real-time parameters are chosen and modulated to stabilize the echo from this point. Recall that specific vectors in this coordinate frame are

$$\begin{aligned} \mathbf{r}_{c,n} &= \text{the vector from the SRP to the radar, and} \\ \mathbf{s} &= \text{the vector from the SRP to the target scatterer point.} \end{aligned} \quad (21)$$

Note that $\mathbf{r}_{c,n}$ is also generally a function of pulse index n . The radar and target vectors identify the following respective Cartesian coordinates

$$\begin{aligned} x_c, y_c, z_c &= \text{the coordinate identified by } \mathbf{r}_{c,n} \text{ in the } x, y, z \text{ coordinate frame, and} \\ s_x, s_y, s_z &= \text{the coordinate identified by } \mathbf{s} \text{ in the } x, y, z \text{ coordinate frame.} \end{aligned} \quad (22)$$

The radar coordinates are related to the angles in Figure 2 as

$$\begin{aligned} x_c &= |\mathbf{r}_{c,n}| \cos \psi_n \sin \alpha_n, \\ y_c &= -|\mathbf{r}_{c,n}| \cos \psi_n \cos \alpha_n, \\ z_c &= |\mathbf{r}_{c,n}| \sin \psi_n, \end{aligned} \quad (23)$$

where for the n th pulse,

$$\begin{aligned}
 \alpha_n &= \text{the aperture angle between the radar position and the center of the} \\
 &\quad \text{synthetic aperture,} \\
 \psi_n &= \text{the grazing angle with respect to horizontal at the SRP,} \\
 \psi_0 &= \text{the nominal grazing angle with respect to horizontal at the SRP,} \\
 &\quad \text{generally expected at the aperture center.}
 \end{aligned} \tag{24}$$

Note that in spite of our neglecting a subscript for these individual Cartesian coordinates (for our convenience), they are still functions of pulse index n .

We note that sampling angles α_n may be specified to offer processing advantages for other modes and algorithms. For example they may be selected based on constant radar Pulse Repetition Frequency (PRF), constant angular increment, or constant increments in some function of the angle such as $\tan \alpha$.

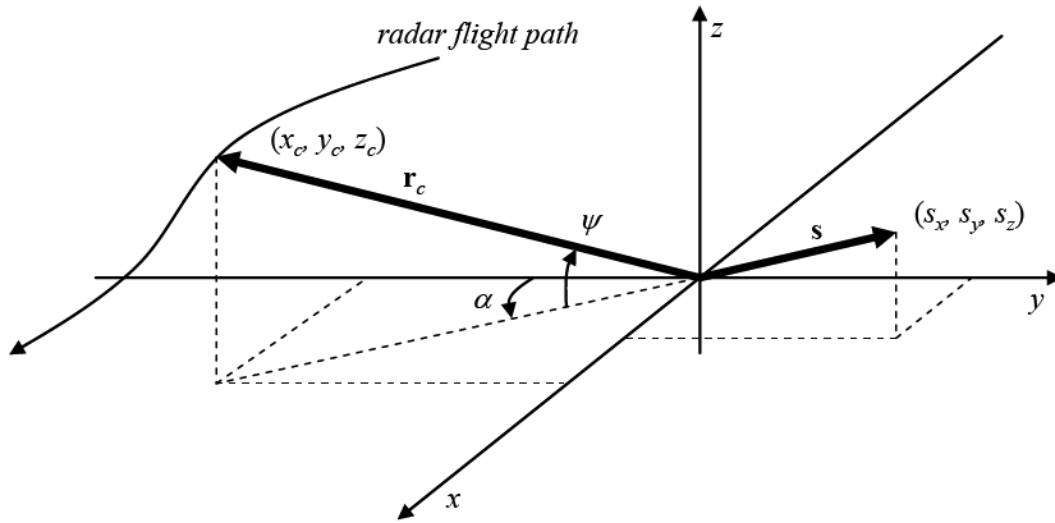


Figure 2. Radar data collection geometry.

We state without elaboration that if the angular spatial sampling rate varies substantially from something nearly linear, then we may need to accommodate what amounts to multi-rate sampling characteristics, and scale the data inversely to the spatial sampling rate. We will ignore this for the remainder of this report.

We also for convenience define the nominal range from the SRP to the center of the synthetic aperture (APC_0) as

$$\mathbf{r}_{c0} = \text{the vector from the SRP to the radar at } APC_0. \tag{25}$$

Furthermore, we define the vector from the target point to the radar as

$$\mathbf{r}_{s,n} = \mathbf{r}_{c,n} - \mathbf{s} = \text{vector from target point to the radar.} \quad (26)$$

We note that in terms of their coordinates

$$\begin{aligned} |\mathbf{r}_{c,n}| &= \sqrt{x_c^2 + y_c^2 + z_c^2}, \\ |\mathbf{s}| &= \sqrt{s_x^2 + s_y^2 + s_z^2}, \text{ and} \\ |\mathbf{r}_{s,n}| = |\mathbf{r}_{c,n} - \mathbf{s}| &= \sqrt{(x_c - s_x)^2 + (y_c - s_y)^2 + (z_c - s_z)^2}. \end{aligned} \quad (27)$$

It becomes useful to identify a slant-range offset as

$$s_{r,n} = |\mathbf{r}_{s,n}| - |\mathbf{r}_{c,n}| = |\mathbf{r}_{c,n} - \mathbf{s}| - |\mathbf{r}_{c,n}|. \quad (28)$$

We note that this definition identifies the range offset as a function of the specific pulse. We further note that the range offset can be approximated as

$$s_{r,n} \approx \begin{pmatrix} -s_x \cos \psi_n \sin \alpha_n \\ +s_y \cos \psi_n \cos \alpha_n \\ -s_z \sin \psi_n \end{pmatrix}, \quad (29)$$

although we will not use this in the image formation algorithm development. It is for informational purposes only. This is sometimes called the “far-field assumption” or “far-field approximation.” We also stipulate that higher-order approximations can be useful to extend image formation performance in some cases.¹⁰

2.3 Waveform Parameter Modulation

The “wavenumber projection compensation factor” κ_n defines the fundamental modulation characteristic for the waveform center frequency and chirp rate on a pulse-to-pulse basis, as a function of the instantaneous radar position. This is in fact a motion compensation factor that facilitates several traditional image formation algorithms, including PFA, OSA, and variants. We make the prescient observation that FBP is ‘not’ facilitated by modulating this factor. Nevertheless, FBP can certainly accommodate the factor, as long as it knows what it is.

Generally, for range-Doppler transform-based SAR algorithms, it is convenient to force

$$\kappa_n = \frac{\cos \psi_0}{\cos \psi_n \cos \alpha_n}. \quad (30)$$

This does require knowledge of an individual pulse's location with respect to an aperture center. However, specifically for VideoSAR, where apertures are overlapped and reuse data, the aperture center is typically determined after the pulse is collected, and in fact the pulse may occupy different positions in different synthetic apertures.

Consequently the waveform parameters may be modulated such that

$$\kappa_n = \frac{\cos\psi_0}{\cos\psi_n}. \quad (31)$$

Of course, these modulations are mere conveniences, particularly for range-Doppler algorithms. Perfectly good SAR data absent these conveniences can be collected without any parameter modulation, that is, with $\kappa_n = 1$. For the remainder of this report, we shall accommodate all collection modes, and carry the general parameter κ_n .

2.4 Echo Timing

The relative delay between scatterer and scene center is a function of data collection geometry, namely

$$\tau_{sc,n} = (t_{s,n} - t_{c,n}) = \frac{2}{c} s_{r,n}. \quad (32)$$

Expanding the geometry terms yields

$$\tau_{sc,n} = \frac{2}{c} \left(\sqrt{(x_c - s_x)^2 + (y_c - s_y)^2 + (z_c - s_z)^2} - \sqrt{x_c^2 + y_c^2 + z_c^2} \right). \quad (33)$$

Recall that it is the variation with index n of the radar position vector $\mathbf{r}_{c,n}$ that defines the synthetic aperture for the data subset of interest.

2.5 The Refined PH Data Model

By operating with a generic κ_n as previously indicated, and substituting Eq. (32) into Eq. (20), the PH data phase model becomes

$$\Phi_V(i, n) = \left(-\frac{2}{c} (\omega_0 + \gamma_0 T_s i) \kappa_n s_{r,n} + \frac{2\gamma_0 \kappa_n}{c^2} (s_{r,n})^2 \right). \quad (34)$$

The first phase term is the principal term of interest for calculating a target's location. The second phase term is often called the Residual Video Phase Error (RVPE), and is an artifact of stretch processing. The model discussed above is just the phase of the scatterer echo. Recall that the actual PH data model is given by

$$x_V(i, n) = A_R e^{j\Phi_V(i, n)}. \quad (35)$$

This amplitude embodies scene reflectivity, antenna beam patterns, range loss, atmospheric loss, and anything that affects the amplitude of a radar echo.

Putting it all together yields the expression for the PH data model for a scatterer at $\mathbf{s}$ as

$$x_V(i, n) = A_R \exp j \left(-\frac{2}{c}(\omega_0 + \gamma_0 T_s i) \kappa_n (|\mathbf{r}_{c,n} - \mathbf{s}| - |\mathbf{r}_{c,n}|) + \frac{2\gamma_0 \kappa_n}{c^2} (|\mathbf{r}_{c,n} - \mathbf{s}| - |\mathbf{r}_{c,n}|)^2 \right). \quad (36)$$

The task at hand is to create a SAR image by estimating the amount of signal reflecting from various probing target positions $\hat{\mathbf{s}}$ (i.e. pixel locations). The location of the actual point target is $\mathbf{s}$, and is embodied in the range differential $s_{r,n}$.

3 Matched Filter Processing

The matched filter is well understood and reported in the literature. Representative texts include those by McDonough, and Whalen,¹¹ and Davenport and Root.¹² It maximizes the Signal-to-Noise Ratio (SNR) of its output in the presence of White Gaussian Noise (WGN). This is what SAR image formation algorithms all try to implement with various trades between image fidelity and processing ease.

Recall from Eq. (36) that our baseband video signal for a single point target scatterer at $\mathbf{s}$ is given by

$$x_V(i, n) = A_R e^{j \left(-\frac{2}{c} (\omega_0 + \gamma_0 T_s i) \kappa_n (|\mathbf{r}_{c,n} - \mathbf{s}| - |\mathbf{r}_{c,n}|) + \frac{2\gamma_0 \kappa_n}{c^2} (|\mathbf{r}_{c,n} - \mathbf{s}| - |\mathbf{r}_{c,n}|)^2 \right)}. \quad (37)$$

For this data we now create a filter function

$$h(i, n, \hat{\mathbf{s}}) = e^{j \left(-\frac{2}{c} (\omega_0 + \gamma_0 T_s i) \kappa_n (|\mathbf{r}_{c,n} - \hat{\mathbf{s}}| - |\mathbf{r}_{c,n}|) + \frac{2\gamma_0 \kappa_n}{c^2} (|\mathbf{r}_{c,n} - \hat{\mathbf{s}}| - |\mathbf{r}_{c,n}|)^2 \right)}. \quad (38)$$

Here we designate a probing test location as

$$\hat{\mathbf{s}} = \text{test location}. \quad (39)$$

A collection of these locations define the image pixels. Note that Eq. (38) is just the expected response from a unit-amplitude signal at the test location.

The matched filter is implemented as a correlation. The output of this filter (for the particular target location being tested) is calculated as

$$y(\hat{\mathbf{s}}) = \sum_i \sum_n \left\{ x_V(i, n) h(i, n, \hat{\mathbf{s}})^* \right\}, \quad (40)$$

where “*” denotes complex conjugate.

Performing this double-summation for every test location $\hat{\mathbf{s}}$ of interest within a 2-D grid yields the SAR image.

A direct implementation of this algorithm is very computationally intensive, and generally prohibitively so. As a result, we attempt to select waveforms and algorithms that get us very close to this result, but with a more tolerable computational load. Nevertheless, virtually all SAR image formation algorithms attempt to at least approximate this “matched-filter” output.

Transforms of the Video Signal

Now suppose that we create a function of the video signal, which we will term a “transform,” defined arbitrarily as

$$F \{x_V(i, n)\} = \text{transform of video signal.} \quad (41)$$

We stipulate that this transform is complete and invertible; something like a Fourier Transform across index i . We recall that the Discrete Fourier Transform (DFT) employs exponential functions to form a complete set of basis functions for the space of periodic functions. In fact, we identify a suitable transform for our purposes as one that projects the PH data onto a new set of orthonormal basis functions, which is exactly what a DFT does.

We can then create a new filter function based on the old one by applying the same transform to the previous filter as

$$F \{h(i, n, \hat{s})\}. \quad (42)$$

This new transformed filter is matched to Eq. (41), the transform of the video signal.

The right transform might cause the data to compress, allowing the replacement of one of the indices with a different one, with the actual target location effectively spanning a shorter domain in the new index k .

Consequently, we may now calculate the image as

$$y(\hat{s}) = \sum_k \sum_n F \{x_V(i, n)\} F \{h(i, n, \hat{s})\}^*, \quad (43)$$

where we can sum over a shorter domain k , with a net savings in computations (in spite of the video signal transform).

This is in fact precisely what the FBP algorithm does, with the transform being one that provides range compression. Details are offered in the following sections.

4 Filtered Backprojection Processing

4.1 Processing Overview

The fundamental processing steps to implement FBP are illustrated in Figure 3. These consist of

PH Data Preprocessing

Image Formation Processing

Image Post-Processing

There is some latitude in assigning specific corrections and processing steps to these major groups. That is, some individual operations might find homes in more than one of these major groupings. For example, a range-loss correction might be applied in the Data Preprocessing step, or might be deferred to the Image Post-Processing step instead.

In addition, some of these major processing steps provide options, depending on the desired fidelity of the result, and processing and memory restrictions. Nevertheless, all these fundamental processing steps will be addressed individually in the subsequent discussion.

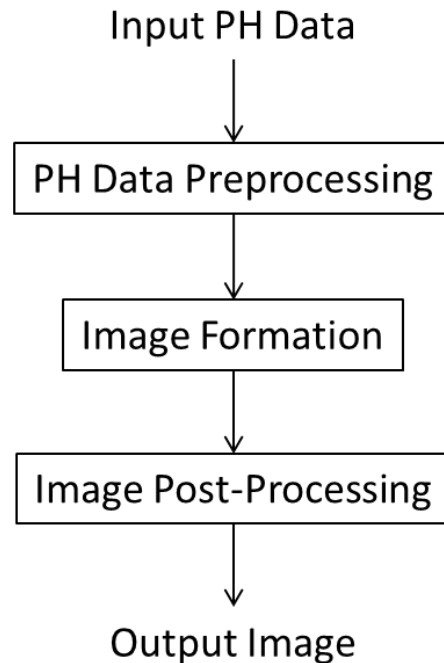


Figure 3. Fundamental steps in the Filtered Backprojection (FBP) algorithm for SAR image formation.

4.2 PH Data Preprocessing

Data preprocessing is comprised of the following principal steps, also illustrated in Figure 4.

1. Processing with a Ramp Filter,
2. Perform Range Compression,
3. RVPE Compensation, and
4. Range-loss Compensation.

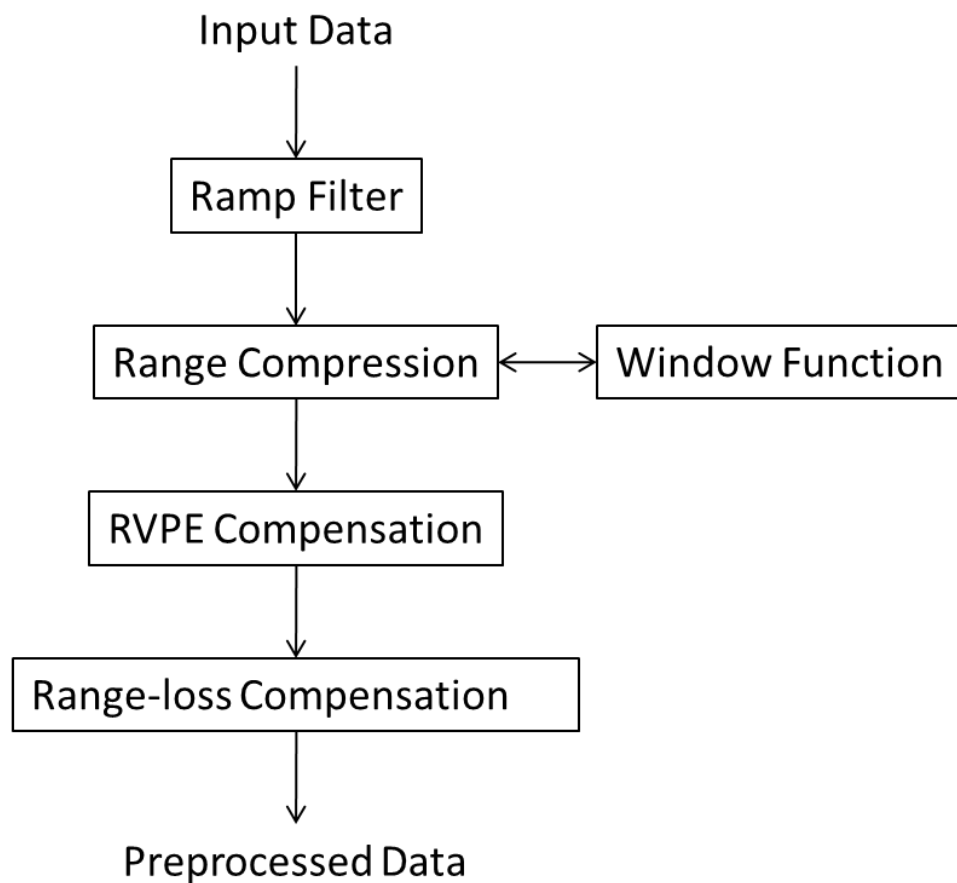


Figure 4. Data preprocessing steps.

For most SAR systems, the percent bandwidth is small enough that the ramp filter can be ignored. It is included here for completeness.

For small scene sizes and long pulses, the RVPE compensation may also sometimes be omitted.

After range compression, we also have an opportunity to compensate for range-loss, that loss in power/energy due to range variations in target locations.

We begin the following discussion by expanding on the PH data model (not just the phase), substituting Eq. (32) into Eq. (36), to write

$$x_V(i, n) = A_R \exp j \left(-\frac{2}{c} (\omega_0 + \gamma_0 T_s i) \kappa_n s_{r,n} + \frac{2\gamma_0 \kappa_n}{c^2} (s_{r,n})^2 \right). \quad (44)$$

We are now set to examine these compensation steps in detail.

4.2.1 Ramp Filter

The origin of this filter is from general Computed Tomography.¹ This filter's purpose is to compensate for the sampling density variations in the wavenumber domain of the target scene. Essentially, at the lower end of the radar band, the samples are closer together than at the higher end of the radar band. This is an artifact of the polar (or trapezoidal) grid locations that the raw PH data samples represent. illustrates the PH data sample positions in the wavenumber domain for typical SAR data.

This is the 'filter' in Filtered Backprojection. If this filtering occurs as a convolution after range compression, then this is the 'convolution' in Convolution Backprojection.

As previously stated, since this filter characteristic is a slowly varying amplitude function, essentially a relatively flat high-pass filter, this filter can be ignored with little detriment to the eventual SAR images. The impact of doing so on the SAR's Impulse Response (IPR) is typically virtually indiscernible. However, we will include it here for completeness.

We take advantage of the fact that our de-chirped data are essentially samples in the wavenumber domain of the target scene being imaged. We then define the filter function as scaling with frequency as projected in a horizontal plane in the wavenumber domain, namely

$$H_{ramp}(i, n) = \frac{(\omega_0 + \gamma_0 T_s i)}{\omega_0} \kappa_n \frac{\cos \psi_n}{\cos \psi_0}, \quad (45)$$

with the stipulation that

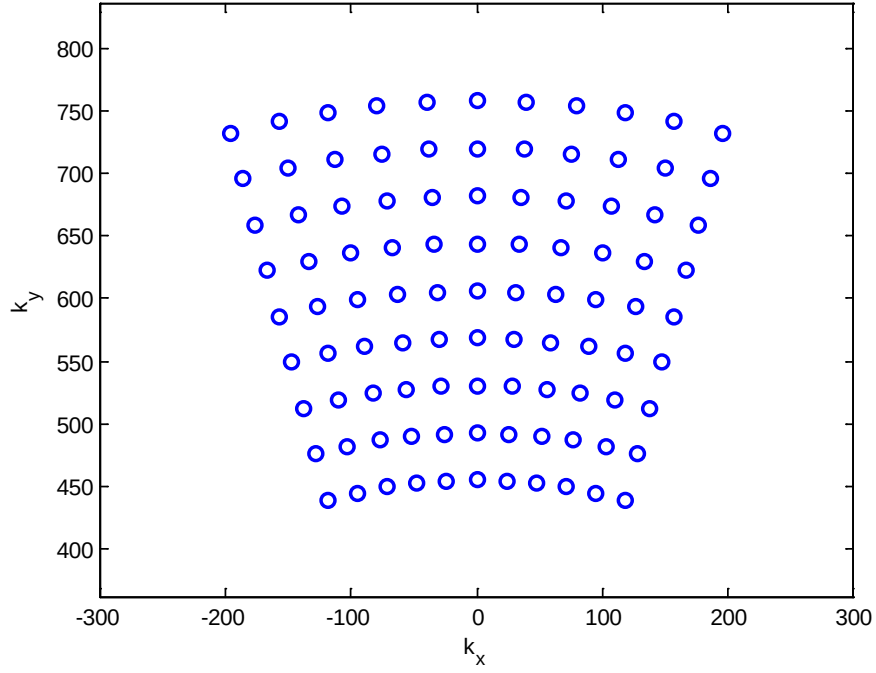


Figure 5. Wavenumber sample locations for typical SAR data. Note how samples towards the bottom of the data array are closer together than samples at the top of the data array. The effect of this is compensated by the “Ramp” filter.

$$(\omega_0 + \gamma_0 T_s i) > 0. \quad (46)$$

The application of this filter is given by

$$x'_V(i, n) = H_{ramp}(i, n) x_V(i, n). \quad (47)$$

We refer the reader to the previously cited reference on Computed Tomography for further insight into the need for, and nature of this filter.

4.2.2 Range Compression

All other subsequent signal processing presumes that the data is range-compressed, that is, each pulse's data has been processed to resolve ranges. This is then the next step in the preprocessing chain. The task is to identify and separate the range offsets $s_{r,n}$.

Towards this end we expand the phase term and rewrite the PH data model in terms of a frequency term of index i , and some phase terms that don't involve index i , namely as

$$x'_V(i, n) = A_R H_{ramp}(i, n) \exp j \left(-\frac{2}{c} \gamma_0 T_s \kappa_n s_{r,n} i - \frac{2}{c} \omega_0 \kappa_n s_{r,n} + \frac{2\gamma_0 \kappa_n}{c^2} (s_{r,n})^2 \right). \quad (48)$$

We note that $s_{r,n}$ is embodied in the frequency term of index i . We will take advantage of this.

To proceed, we first define for this report the forward and reverse Discrete Fourier Transform (DFT) pair as

$$\begin{aligned} Z(v) &= DFT_{V,i} \{z(i)\} = \sum_{i=-I/2}^{I/2-1} z(i) \exp \left\{ -j \frac{2\pi v}{V} i \right\}, \text{ and} \\ z(i) &= IDFT_{V,v} \{Z(v)\} = \frac{1}{V} \sum_{v=-V/2}^{V/2-1} Z(v) \exp \left\{ j \frac{2\pi i}{V} v \right\}, \end{aligned} \quad (49)$$

with integer indices

$$\begin{aligned} -I/2 \leq i < I/2, \text{ and} \\ -V/2 \leq v < V/2, \text{ with } V \text{ being the number of DFT output samples.} \end{aligned} \quad (50)$$

We pick these indexing limits for convenience to this development, recognizing that most software tends to like indices that start at 0 or 1 and increase from there. There are straightforward conversions from our indexing scheme to others that may also involve a constant phase correction.

Substituting $z(i) = \exp \{j\omega i\}$ into (49), we identify the DFT of a complex sinusoid with respect to index i as the specific function

$$DFT_{V,i} \{ \exp \{j\omega i\} \} = W_I \left(\frac{I}{2\pi} \left(\omega - \frac{2\pi}{V} v \right) \right), \quad (51)$$

where in this case we identify the expanded function form as

$$W_M(f) = M \left(\frac{\sin(\pi f)}{M \sin(\pi f/M)} \right) \exp \left\{ -j \frac{\pi f}{M} \right\}. \quad (52)$$

Note that this is merely the DFT of a sampled rectangle, or Boxcar, function, implicit with the untapered unit modulus on the left side of Eq. (51). Of significance, this DFT result peaks strongly at output index value $v = \omega(V/2\pi)$. This is the Impulse Response (IPR) of an untapered complex sinusoid. Later we will redefine and generalize the IPR

function $W_I(\cdot)$ for a tapered complex sinusoid, but retain the characteristics of peaking at $v = \omega(V/2\pi)$.

Using this, we take the forward transform of the PH model with respect to index i and define the result as

$$G(v, n) = DFT_{V,i} \{x'_V(i, n)\} \approx \left(A_R \exp j \left(-\frac{2}{c} \omega_0 \kappa_n s_{r,n} + \frac{2\gamma_0 \kappa_n}{c^2} (s_{r,n})^2 \right) \right) \times W_I \left(\frac{I}{2\pi} \left(-\frac{2}{c} \gamma_0 T_s \kappa_n s_{r,n} - \frac{2\pi}{V} v \right) \right). \quad (53)$$

We have essentially ignored the effect of $H_{ramp}(i, n)$ in this output, as the fairly shallow amplitude ramp that this filter represents has effectively very small impact to the shape or phase of $G(v, n)$. Basically, while this filter is a big deal for medical imaging, it is not so much for typical SAR imaging.

Nevertheless, based on the characteristics of Eq. (52), and how they relate to the arguments in the DFT result in Eq. (53), we identify that there now exists a correspondence between index v and parameter $s_{r,n}$. Specifically, we can calculate this from the peak of the IPR function as

$$s_{r,n} = - \left(\frac{2\pi}{V} \right) \left(\frac{c}{2\gamma_0 T_s \kappa_n} \right) v_{peak}. \quad (54)$$

Before we move on to subsequent processing, we first investigate some finer points of range compression.

4.2.2.1 Window Functions for Sidelobe Control

Note that the DFT of a complex sinusoid in the previous section has rather undesirable sidelobe levels. More commonly, window functions are employed that taper the data prior to the DFT for sidelobe control. A common window function for SAR processing might be a -35 dB Taylor window (with $\bar{n} = 4$). Other window taper functions such as Hamming, or Hanning windows are also frequently used, and are illustrated in Figure 6.

We now define a generic window function as

$$w_M(m) = \text{window function, } -M/2 \leq m < M/2, \quad (55)$$

for some window length M , and identify its DFT as the re-designated function

$$W_M(f) = DFT_{V,m}(w_M(m)) = \text{the IPR of the window function.} \quad (56)$$

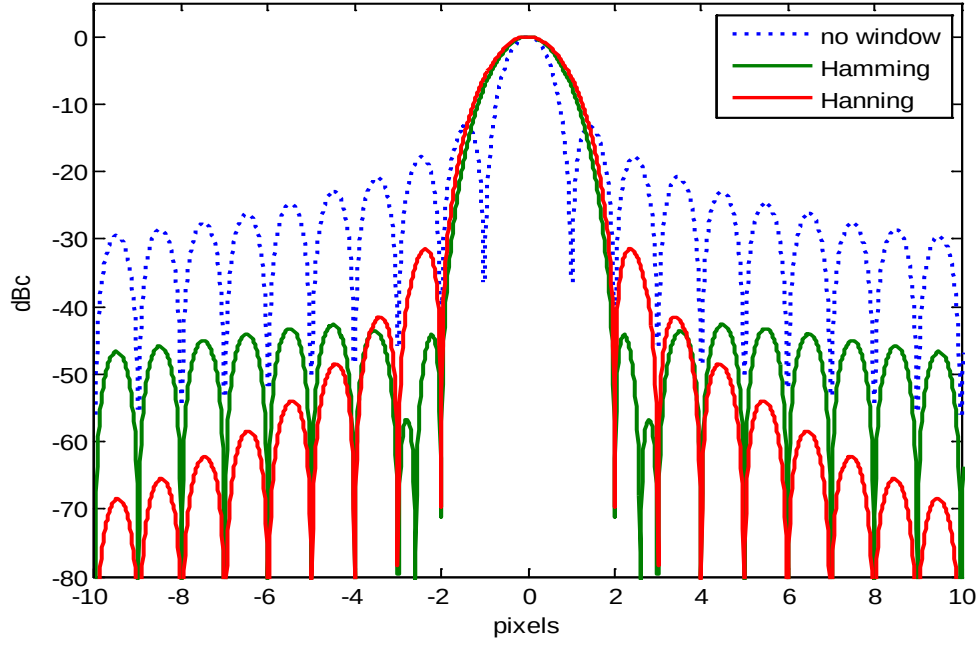


Figure 6. Window IPR plots. Popular window functions include Hamming, Hanning, Taylor, and Blackman. There are many others. The cost is a wider mainlobe. This is generally considered a good trade.

For a rectangular, or “boxcar,” window, this is consistent with Eq. (52). We will also generally assume (or insist) that the DC gain of the window function is M . What this really means is that if we sum all the window function weights, the result is M .

A range-processing DFT that employs window functions on the data will yield

$$G(v, n) = DFT_{V,i} \{ w_I(i) x'_V(i, n) \}, \quad (57)$$

which for the simple model of the previous section can be expanded to

$$G(v, n) = A_R \left[W_I \left(\frac{I}{2\pi} \left(-\frac{2}{c} \gamma_0 T_s \kappa_n s_{r,n} - \frac{2\pi}{V} v \right) \right) \times \exp j \left(-\frac{2}{c} \omega_0 \kappa_n s_{r,n} + \frac{2\gamma_0 \kappa_n}{c^2} (s_{r,n})^2 \right) \right]. \quad (58)$$

This is just a rearranged version of Eq. (53), albeit with a generalized definition of the IPR function $W_I(\cdot)$. This can be expressed in terms of resolution and pixel spacing as

$$G(v, n) = A_R \left(W_I \left(\frac{a_{wr}}{\rho_r} (-\kappa_n s_{r,n} - \delta_r v) \right) \times \exp j \left(-\frac{2}{c} \omega_0 \kappa_n s_{r,n} + \frac{2\gamma_0 \kappa_n}{c^2} (s_{r,n})^2 \right) \right), \quad (59)$$

where

$$\begin{aligned} \rho_r &= \frac{a_{wr} c}{2B_{eff}} = \text{nominal slant-range resolution,} \\ \delta_r &= \frac{\rho_r}{a_{osr}} = \text{nominal slant-range pixel spacing,} \end{aligned} \quad (60)$$

and these in turn depend on

$$\begin{aligned} B_{eff} &= \gamma_0 T_s I / 2\pi = \text{effective bandwidth of the data,} \\ a_{osr} &= \frac{V}{I} a_{wr} = \text{IPR oversampling factor.} \end{aligned} \quad (61)$$

An important point here is that this is the nominal slant-range resolution. Furthermore, we stress that the effective bandwidth B_{eff} is the bandwidth that manifests in the PH data, which may be somewhat less than the bandwidth of the actual TX pulse.

With respect to following discussions, the window function is applied over only the data, regardless of the DFT length, as opposed to over perhaps the entire length of a zero-padded data vector.

Alternatively, we may write Eq. (59) as

$$G(v, n) = A_R \left(W_I \left(-\frac{a_{wr}}{\rho_r} \kappa_n (s_{r,n} - \hat{s}_{r,n}) \right) \times \exp j \left(-\frac{2}{c} \omega_0 \kappa_n s_{r,n} + \frac{2\gamma_0 \kappa_n}{c^2} (s_{r,n})^2 \right) \right), \quad (62)$$

where we are able to correspond slant-range offset from the SRP to the index value as

$$\hat{s}_{r,n} = -\left(\frac{2\pi}{V} \right) \left(\frac{c}{2\gamma_0 T_s \kappa_n} \right) v = -\frac{\delta_r}{\kappa_n} v. \quad (63)$$

4.2.2.2 Using Power of 2 DFT

We have developed the range compression in terms of a generalized DFT. In practice, we usually wish to use an optimized (from a computation-time standpoint) version of this known as the Fast Fourier Transform (FFT). Most often, the FFT requires input and output vectors to be equal, and furthermore to be a power of two, although other radix fast algorithms sometimes are available.

In the development above we are beginning with data indexed by i , with a vector length I . If I is not a power of 2, then in order to employ the FFT, we need to zero-pad (add samples of zero), to increase the vector length to the FFT size, namely V , which is a power of 2. Of course, this implies

$$V \geq I. \quad (64)$$

With the DFT defined as above, the zero-padding is done evenly split between prepending and appending the zero values. We stress that where we add the zeros is extremely important.

4.2.2.3 Up-Sampling the Range-Compressed Data

We observe that since we transform to the Fourier domain with respect to index i , we have an opportunity to aid subsequent processing by synthetically increasing the sampling rate of the data in the transformed data described by $G(v, n)$. Recall that the IPR oversampling factor was defined as

$$a_{osr} = \frac{\rho_r}{\delta_r} = \frac{V}{I} a_{wr}. \quad (65)$$

Note that as the DFT length V increases for a fixed data vector length I , the oversampling factor increases correspondingly, and the output pixel spacing gets finer proportionately. Essentially, ever finer interpolation of the output data occurs, albeit limited to specific rational factors of the resolution.

Later, during image formation itself, we will need to interpolate this data to arbitrary locations, not limited to rational factors. This will be considerably easier (less complicated) with larger oversampling factors in the range compressed data.

As discussed in the previous section, the data needs zeros both prepended and appended to an overall length V that is generally larger than the data vector length I . It is important for our data model, that a key to making this work correctly with our indexing scheme is that the zeros need to be added equally on either side of the data vector. If the zeros are added in unequal amounts, then additional phase corrections may need to be applied in the processing. This is quite doable, but not addressed in this report.

Looking ahead, this up-sampling operation reduces the fractional bandwidth of our signal with respect to the output spatial sampling frequency. This will make resampling and interpolation easier during image formation.

4.2.2.4 Some Comments

We note that this report employed the forward DFT for range compression. With no loss of effectiveness, it could just as easily have employed the Inverse-DFT (IDFT) instead. The only consequence of using an IDFT instead of the DFT is a sign change on the output index value, and an amplitude adjustment. Consequently, using one is no more ‘right’ than the other.

In addition, other pre-DFT correction vectors can be combined with the window function into a single multi-purpose correction. For example, the ramp filter may be combined with the window function into a single pre-computed correction vector.

4.2.3 **RVPE Compensation**

Repeating Eq. (62) for convenience, we recall the range compressed signal is given by

$$G(v, n) = A_R \left(W_I \left(-\frac{a_{wr}}{\rho_r} \kappa_n (s_{r,n} - \hat{s}_{r,n}) \right) \times \exp j \left(-\frac{2}{c} \omega_0 \kappa_n s_{r,n} + \frac{2\gamma_0 \kappa_n}{c^2} s_{r,n}^2 \right) \right). \quad (66)$$

Note that the phase terms contain a dependence on $s_{r,n}$. The first phase term, that which is linear in $s_{r,n}$, is responsible for azimuth defocusing, as well as range walk. This contains the Doppler frequency term that is exploited for azimuth resolution. Its compensation is essentially focusing the data. We will deal with it in the image formation stage.

The second phase term, that which contains $s_{r,n}^2$, is called the “Residual Video Phase Error” (RVPE), or sometimes “skew.” It is an artifact of stretch processing. Its compensation is sometimes termed “deskewing,” and can be accomplished in this preprocessing stage.

Note that for each pulse n , the model $G(v, n)$ is a vector where each index v corresponds to a particular range offset $\hat{s}_{r,n}$. This lets us construct the first, RVPE, correction vector

$$X_{correction,1}(v, n) = \exp j \left\{ -\frac{2\gamma_0 \kappa_n}{c^2} \hat{s}_{r,n}^2 \right\}, \quad (67)$$

which we can expand explicitly in terms of the range index value to

$$X_{correction,1}(v, n) = \exp j \left\{ - \left(\frac{1}{2\kappa_n \gamma_0 T_s^2} \right) \left(\frac{2\pi}{V} \right)^2 v^2 \right\}. \quad (68)$$

The application of the correction is described by the multiplication

$$G'(v, n) = G(v, n) X_{correction,1}(v, n). \quad (69)$$

This yields the compensated range-compressed data model as simply

$$G'(v, n) = A_R \left[W_I \left(-\frac{a_{wr}}{\rho_r} \kappa_n (s_{r,n} - \hat{s}_{r,n}) \right) \exp j \left(-\frac{2}{c} \omega_0 \kappa_n s_{r,n} \right) \right. \\ \left. \times \exp j \left(\frac{2\gamma_0 \kappa_n}{c^2} (s_{r,n}^2 - \hat{s}_{r,n}^2) \right) \right]. \quad (70)$$

In the vicinity of the mainlobe of the compensated compressed range response, the second phase term is very small, and can be hereafter ignored. The RVPE phase error has therewith been compensated. Consequently, the compensated range-compressed data model becomes

$$G'(v, n) = A_R W_I \left(-\frac{a_{wr}}{\rho_r} \kappa_n (s_{r,n} - \hat{s}_{r,n}) \right) \exp j \left(-\frac{2}{c} \omega_0 \kappa_n s_{r,n} \right). \quad (71)$$

We like this. This data model is now ready for the next processing steps.

4.2.4 Compensating for Range-Loss

Up to now, post range compression, we have dealt with phase errors. We note that with the range compressed signal we also have the opportunity to correct for some amplitude errors as well. Recall that after the phase error corrections, we have the range-compressed data model

$$G'(v, n) = A_R W_I \left(-\frac{a_{wr}}{\rho_r} \kappa_n (s_{r,n} - \hat{s}_{r,n}) \right) \exp j \left(-\frac{2}{c} \omega_0 \kappa_n s_{r,n} \right). \quad (72)$$

This affords us an opportunity to correct some amplitude effects.

4.2.4.1 Range-Loss due to Spherical Wavefronts

From an analysis of the radar equation,¹³ we are given that radar echo power diminishes as the fourth power of range for an individual pulse, due to spherically expanding wavefronts. This means that the signal magnitude (being proportional to the square-root

of power) diminishes as the square of range. This can be compensated, or normalized with respect to the SRP by applying the second correction

$$X_{correction,2}(v,n) = \left(\frac{|\mathbf{r}_{c,n}| + \hat{s}_{r,n}}{|\mathbf{r}_{c,n}|} \right)^2. \quad (73)$$

With this correction the model now incorporates an altered amplitude and becomes

$$G''(v,n) = X_{correction,2}(v,n)G'(v,n), \quad (74)$$

which can be expanded to

$$G''(v,n) = A'_R W_I \left(-\frac{a_{wr}}{\rho_r} \kappa_n (s_{r,n} - \hat{s}_{r,n}) \right) \exp j \left(-\frac{2}{c} \omega_0 \kappa_n s_{r,n} \right), \quad (75)$$

where the range-loss correction is now embodied in the new amplitude factor A'_R .

4.2.4.2 Range-Loss due to Antenna Pattern

In addition to the loss due to spherically expanding wavefronts, the antenna beam itself imparts a relative loss with respect to its boresight direction. We often model the antenna beam pattern loss as independent elevation pattern loss and azimuth pattern loss. For a flat target scene, the elevation pattern loss can be modeled as a relatively straightforward range loss in addition to the range loss previously discussed. Azimuth pattern loss is addressed later in section 4.4.1.

Of course, the nature of the antenna pattern loss is dependent on just where the antenna is pointed. While it is customary to point the antenna boresight at the SRP, this need not be the case. However, for the subsequent development below, we shall indeed presume that the antenna is pointed at the SRP.

We define the 2-way signal relative magnitude as function of elevation offset

$$G_{el,ant}(\phi_{el}) = \text{2-way signal magnitude gain}, \quad (76)$$

where ϕ_{el} = the elevation angle offset from boresight. In this development $G_{el,ant}(\phi_{el})$ is positive, real, and no greater than unity. In the direction of boresight $G_{el,ant}(0) = 1$.

We identify the elevation offset for a flat target scene as

$$\phi_{el} = \psi_n - \text{asin} \left(\frac{|\mathbf{r}_{c,n}|}{|\mathbf{r}_{c,n}| + \hat{s}_{r,n}} \sin \psi_n \right). \quad (77)$$

This is not quite true for a non-flat target scene, but is still often “good enough.” Nevertheless, for a flat target scene, we would instead define the earlier second correction to include this correction as

$$X_{\text{correction},2}(v,n) = \frac{\left(\frac{|\mathbf{r}_{c,n}| + \hat{s}_{r,n}}{|\mathbf{r}_{c,n}|} \right)^2}{G_{el,ant} \left(\psi_n - \text{asin} \left(\frac{|\mathbf{r}_{c,n}|}{|\mathbf{r}_{c,n}| + \hat{s}_{r,n}} \sin \psi_n \right) \right)}. \quad (78)$$

With this adjustment, the processing could proceed as before.

4.2.4.3 Some Comments

As before, several of these post-DFT corrections might be combined into a single correction vector. For example the RVPE correction might be combined with the range-loss and antenna elevation pattern correction into a single correction vector.

4.2.5 **Alternative Waveforms and Processing**

We have arrived at a PH data model given in Eq. (75) assuming stretch processing of an LFM chirp waveform. However, there is nothing in Eq. (75) that is any longer unique to stretch processing of an LFM chirp waveform. That is, other waveforms with a more direct matched-filter or correlation processing can get us to the same expression.

Hereafter in this report, the particular waveform and specific range-compression technique employed become irrelevant. All we require is that Eq. (75) adequately expresses the range-compressed result.

4.3 Image Formation

We begin by repeating the PH data model of the previous section, as a result of preprocessing, here as

$$G''(v, n) = A'_R W_I \left(-\frac{a_{wr}}{\rho_r} \kappa_n (s_{r,n} - \hat{s}_{r,n}) \right) \exp j \left(-\frac{2}{c} \omega_0 \kappa_n s_{r,n} \right). \quad (79)$$

We now wish to create an ‘image’ of this data. That is, we wish to see how this target echo response manifests in a 2-D array of image sample locations.

We do so by continuing with our FBP algorithm. We have done the “filtering” part of the algorithm, as well as other preprocessing steps. Now we need to perform the “backprojection” part of the algorithm.

The essence of FBP image formation is now the following backprojection procedure, also illustrated in Figure 7.

1. Create a grid of image sample locations, and
2. Process the PH data to the image sample grid, one pulse at a time.

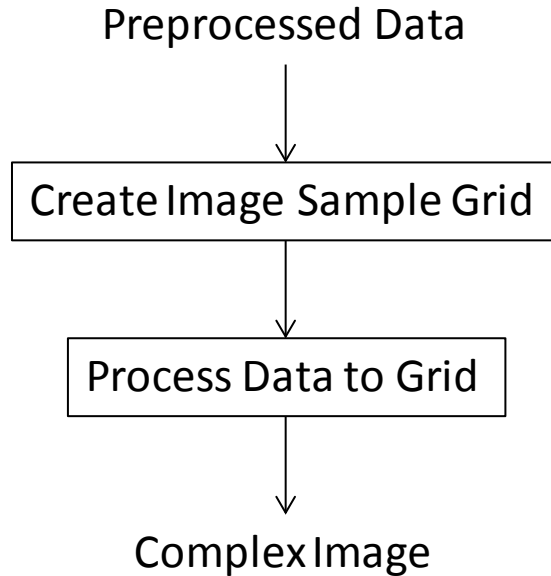


Figure 7. Basic Backprojection procedure.

The processing itself requires that for each pulse of collected data n , and for each sample pixel location in the image sample grid, we apply the following operations, also illustrated in Figure 8.

- Calculate range difference between image sample location and SRP,
- Calculate the fractional range difference index v' ,
- Interpolate the PH vector to the fractional range difference index v' ,
- Compensate for the Doppler phase term, and
- Accumulate the interpolated value into the target array.

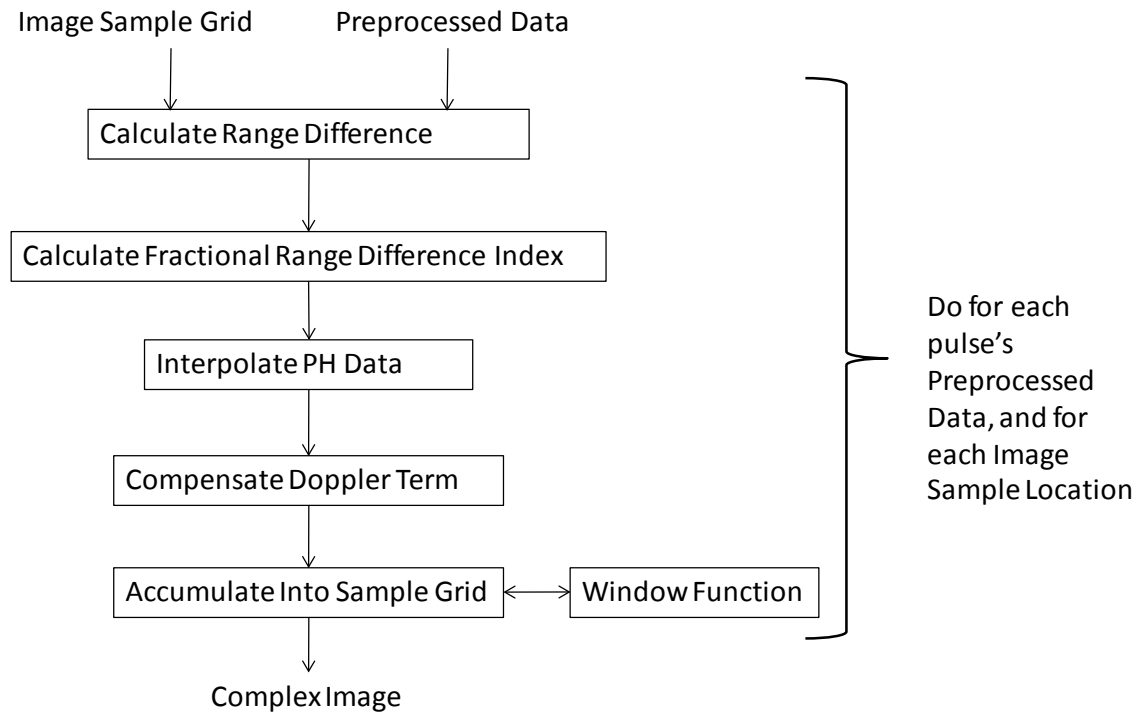


Figure 8. Detailed Backprojection processing steps.

4.3.1 Creating the Image Sample Grid

We shall presume to form a typical SAR image that is a 2-D map of radar reflectivity. We shall further presume that the image is a rectangular grid of sample locations centered on the SRP.

Neither of these presumptions is mandatory for FBP image formation, but both are nevertheless convenient for us, and quite typical.

In any case, we presume that the image rectangular grid array may be expressed with

$$\begin{aligned}\hat{s}_x &= u_x \delta_x = x\text{-dimension offset, and} \\ \hat{s}_y &= u_y \delta_y = y\text{-dimension offset.}\end{aligned}\tag{80}$$

where

$$\begin{aligned}u_x &= x\text{-dimension index, with } -U_x/2 \leq u_x < U_x/2, \\ u_y &= y\text{-dimension index, with } -U_y/2 \leq u_y < U_y/2, \\ \delta_x &= x\text{-dimension pixel spacing, and} \\ \delta_y &= y\text{-dimension pixel spacing.}\end{aligned}\tag{81}$$

The image size in the ground-plane would then have dimensions

$$\begin{aligned}D_x &= U_x \delta_x = x\text{-dimension image size, and} \\ D_y &= U_y \delta_y = y\text{-dimension image size.}\end{aligned}\tag{82}$$

With knowledge of the local topography, for example with Digital Terrain Elevation Data (DTED), we may have the possibility of using a z-dimension offset for each image pixel. That is, we might allow calculating

$$\hat{s}_z = DTED(\hat{s}_x, \hat{s}_y) = z\text{-dimension offset.}\tag{83}$$

Now, with each index pair (u_x, u_y) we can identify a unique 3-D location $(\hat{s}_x, \hat{s}_y, \hat{s}_z)$.

4.3.2 Process the PH data to the Image Sample Grid

In this section we step through the process displayed in Figure 8. That is, for each preprocessed received echo pulse data with index n , we engage the following steps for each image location index (u_x, u_y) .

4.3.2.1 Calculate Range Difference Between Image Sample Location and SRP

Recall that with each index pair (u_x, u_y) we can identify a unique 3-D location $(\hat{s}_x, \hat{s}_y, \hat{s}_z)$. Therewith we are able to calculate the range to the radar at pulse index n , and in particular wish to calculate the range difference between the image sample location and the SRP as

$$\hat{s}_{r,n} = \sqrt{(x_c - \hat{s}_x)^2 + (y_c - \hat{s}_y)^2 + (z_c - \hat{s}_z)^2} - \sqrt{x_c^2 + y_c^2 + z_c^2}. \quad (84)$$

While this expression is exact, we note that under some conditions (which we will not elaborate) we may approximate this with a ‘far-field approximation’, and calculate

$$\hat{s}_{r,n} \approx -\hat{s}_x \cos \psi_n \sin(\alpha_n + \alpha_{ref}) + \hat{s}_y \cos \psi_n \cos(\alpha_n + \alpha_{ref}) - \hat{s}_z \sin \psi_n, \quad (85)$$

where

$$\alpha_{ref} = \text{rotation angle of the image grid}. \quad (86)$$

This rotation angle is with respect to a projection onto the ground of the vector from APC_0 to the SRP.

We stress that because this is an approximation, using it will manifest errors in the image that become increasingly apparent as scene diameter increases and resolution becomes finer, among other things. We will hereafter proceed with the exact expression.

4.3.2.2 Calculate the Fractional Range Difference Index v'

With a range difference $\hat{s}_{r,n}$ corresponding to our desired pixel location, we can now calculate the corresponding non-integer effective index value as

$$v' = \frac{-\kappa_n \hat{s}_{r,n}}{\delta_r}. \quad (87)$$

Of course this also means that

$$v' \delta_r = -\kappa_n \hat{s}_{r,n}. \quad (88)$$

4.3.2.3 Interpolate the PH Vector to the Fractional Range Difference Index v'

This step essentially accommodates range migration. Given the preprocessed data vector $G''(v, n)$ over index values v , we wish to calculate the interpolated value $G''(v', n)$ at a specific effective index value v' , which results in the data model given by

$$G''(v', n) = A'_R W_I \left(-\frac{a_{wr}}{\rho_r} \kappa_n (s_{r,n} - \hat{s}_{r,n}) \right) \exp j \left(-\frac{2}{c} \omega_0 \kappa_n s_{r,n} \right). \quad (89)$$

Note that for any one pulse n , and any one image location (u_x, u_y) , the entity $G''(v', n)$ is just one complex number.

SAR data interpolation is a rich topic. There is definitely a science to it, meaning that interpolation algorithm details need to be chosen with purpose, else undesired artifacts in the image are likely. A thorough discussion of interpolation is beyond the scope of this report, but good insight can be had in a paper by Doerry, et al.¹⁴

4.3.2.4 Compensate for the Doppler Phase Term

This step focuses the data to the location $(\hat{s}_x, \hat{s}_y, \hat{s}_z)$. We do this by correcting the phase of $G''(v', n)$ by multiplying this particular sample as follows.

$$G'''(v', n) = G''(v', n) \exp j \left(+\frac{2}{c} \omega_0 \kappa_n \hat{s}_{r,n} \right), \quad (90)$$

where recall that $\hat{s}_{r,n}$ corresponds to a particular pixel location $(\hat{s}_x, \hat{s}_y, \hat{s}_z)$. Multiplying this out yields the model

$$G'''(v', n) = A'_R W_I \left(-\frac{a_{wr}}{\rho_r} \kappa_n (s_{r,n} - \hat{s}_{r,n}) \right) \exp j \left(-\frac{2}{c} \omega_0 \kappa_n (s_{r,n} - \hat{s}_{r,n}) \right). \quad (91)$$

Note that this has non-zero amplitude with constant phase over pulse index n only when $\hat{s}_{r,n} = s_{r,n}$. This suggests a global peak response at the unique image location given at some index pair (u_x, u_y) corresponding to location $(\hat{s}_x, \hat{s}_y, \hat{s}_z)$.

4.3.2.5 Accumulate the Interpolated Value Into the Target Array

This is the step that adds this data into the image array at location (u_x, u_y) . This addition is done for all pulses. That is, we calculate the response at a particular pixel as

$$y(u_y, u_x) = \sum_n G'''(v', n), \quad (92)$$

noting that v' itself will vary with pulse index n , and pixel location (u_x, u_y) .

So, when all is said and done, the image formation process is the accumulation of data into the image grid, over all pulses and for all image sample grid locations. The resulting array $y(u_y, u_x)$ is the complex SAR image. This process is illustrated in Figure 9.

Window Functions for Sidelobe Control

We note that in the previous accumulation of processed data into the target array, that all pulses contributed equally to the accumulation. This will result in a characteristic sinc() function in the IPR, with the attendant generally unacceptable sidelobes. To reduce sidelobes, we need to employ a weighted summation, with a taper on the data over pulse index n .

$$y(u_y, u_x) = \sum_n w_N(n) G'''(v', n). \quad (93)$$

We state without elaboration that for large range dimensions, we may on occasion wish to equalize azimuthal resolutions as a function of range by adjusting window parameters as a function of range. We would then adjust the image formation processing operation to

$$y(u_y, u_x) = \sum_n H_w(v', n) G'''(v', n), \quad (94)$$

where

$$H_w(v', n) = \text{two-dimensional tapering function}. \quad (95)$$

Operationally, we might specify the desired resolution at far-range, and then adjust azimuthal window function parameters to effectively reduce Doppler bandwidth for nearer ranges.

4.3.2.6 The Image Model

We need to remember that directions of indices (u_y, u_x) are not necessarily aligned with those of indices (v', n) . There may be a substantial rotation between the two sets of indices.

We can however gain some insight into the nature of the image by expanding the result above to

$$y(u_y, u_x) = \sum_n \left(w_N(n) A'_R W_I \left(-\frac{a_{wr}}{\rho_r} \kappa_n(s_{r,n} - \hat{s}_{r,n}) \right) \times \exp j \left(-\frac{2}{c} \omega_0 \kappa_n(s_{r,n} - \hat{s}_{r,n}) \right) \right). \quad (96)$$

Remember that we are not engaging any processing steps here, merely performing some mathematical operations to understand the nature of the result.

For illustrative purposes we shall presume in this section that $s_z = 0$. In addition, we now make use of some earlier results. We recall that we might approximate

$$s_{r,n} \approx \begin{pmatrix} -s_x \cos \psi_n \sin(\alpha_n + \alpha_{ref}) \\ +s_y \cos \psi_n \cos(\alpha_n + \alpha_{ref}) \end{pmatrix} = \begin{pmatrix} -s'_x \cos \psi_n \sin \alpha_n \\ +s'_y \cos \psi_n \cos \alpha_n \end{pmatrix}, \quad (97)$$

where we have used rotated coordinates for our point reflector defined as

$$\begin{aligned} s'_x &= s_x \cos \alpha_{ref} + s_y \sin \alpha_{ref}, \text{ and} \\ s'_y &= -s_x \sin \alpha_{ref} + s_y \cos \alpha_{ref}. \end{aligned} \quad (98)$$

Expanding a little further, we can calculate the approximation

$$\kappa_n s_{r,n} \approx \cos \psi_0 (-s'_x d\alpha_n + s'_y). \quad (99)$$

This lets us then approximate the image as

$$y(u_y, u_x) \approx \sum_n \left(A'_R W_I \left(\frac{a_{wr}}{\rho_r} (-s'_y \cos \psi_0 - \hat{s}_{r,n}) \right) \times w_N(n) \exp j \left(\frac{2}{c} \omega_0 \cos \psi_0 s'_x d\alpha_n \right) \times \exp j \left(\frac{2}{c} \omega_0 (-\cos \psi_0 s'_y - \hat{s}_{r,n}) \right) \right). \quad (100)$$

By performing the summation, this can be expanded out to

$$y(u_y, u_x) \approx \begin{pmatrix} A'_R W_I \left(\frac{a_{wr}}{\rho_r} (-s'_y \cos \psi_0 - \hat{s}_{r,n}) \right) \\ \times W_N \left(\frac{a_{wa}}{\rho_{x'}} (s'_x - \hat{s}'_x) \right) \\ \times \exp j \left(\frac{2}{c} \omega_0 (-\cos \psi_0 s'_y - \hat{s}_{r,n}) \right) \end{pmatrix}, \quad (101)$$

where the azimuth IPR function contains

$$\begin{aligned} \rho_{x'} &= \frac{a_{wa} \lambda_0}{2Nd\alpha \cos \psi_0} = \text{azimuth resolution, and} \\ \hat{s}'_x &= \text{azimuth offset of image sample location.} \end{aligned} \quad (102)$$

For our immediate purposes we are going to ignore the remaining phase terms, which yields

$$y(u_y, u_x) \approx A'_R W_I \left(\frac{a_{wr}}{\rho_r} (-s'_y \cos \psi_0 - \hat{s}_{r,n}) \right) W_N \left(\frac{a_{wa}}{\rho_{x'}} (s'_x - \hat{s}'_x) \right). \quad (103)$$

From this we observe the following.

- Range sidelobes will extend in the direction of s'_y .
- Azimuth sidelobes will extend in the direction of s'_x .
- In the actual image, the sidelobes will be rotated by an angle of α_{ref} . Consequently, sidelobes will not necessarily be up/down, or sideways.
- In the actual image, nominal range and azimuth resolution will also not necessarily correspond to up/down, or sideways.

These observations will have profound implications in some post-processing operations.

Furthermore, because it is still true that the raw PH data represents an ‘aperture’ in Fourier space¹⁵, and because the Fourier Transform preserves angles, a rotation of the image represents a rotation of the aperture in Fourier space. This has further implications on some post-processing operations, especially those that involve returning to the Fourier space of the target image.

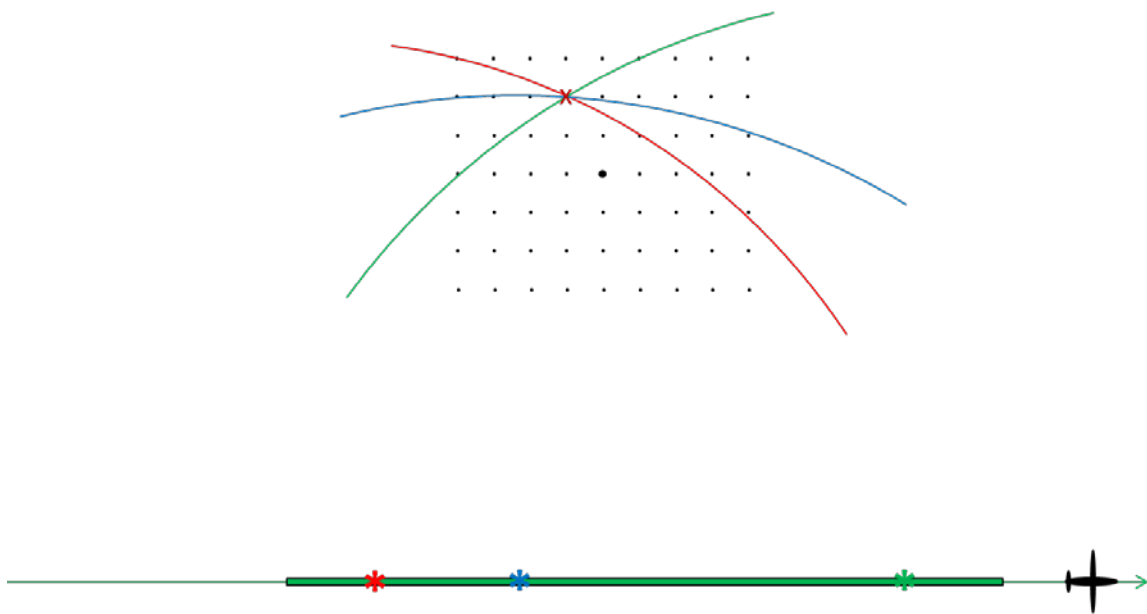


Figure 9. The data from any one pulse, at any one range, is a superposition of all targets along that constant-range arc. It is the different aspect angles that allow separation of targets along that arc. The basic idea is to 1) select a pixel location, 2) interpolate the data to the exact range of the pixel, 3) align the phases, and 4) accumulate the result.

4.4 Image Post-Processing

There are a number of operations that can be applied to the complex image after its formation. We discuss some of these here.

4.4.1 Amplitude Corrections

While focusing a SAR image is principally an exercise in ‘getting the phase right’, we also acknowledge a need to address the amplitude characteristics of the data. Two principal sources of undesirable amplitude fluctuations are

1. Power loss as a function of range due to the expansion of the spherical wave fronts, and
2. Power loss as a function of antenna beam pattern effects.

A third source might be various filter characteristics, such as IF and video filters, and even interpolation filters. We will ignore these here.

We have already addressed compensation for range-dependent loss factors during preprocessing of the data. These range-dependent operations need to occur after range compression. If they are not done elsewhere, then they can be performed as image post-processing operations.

In addition, compensation for azimuth-dependent loss factors, such as the azimuth antenna beam pattern, are dependent on image pixel spatial position, which for FBP means corrections are seemingly best applied after image formation, with the best opportunity being as an image post-processing operation, too.

We define the 2-way signal relative magnitude of the antenna beam taper as a function of azimuth offset with

$$G_{az,ant}(\phi_{az}) = \text{2-way signal magnitude gain}, \quad (104)$$

where ϕ_{az} = the azimuth angle offset from boresight. In this development $G_{az,ant}(\phi_{az})$ is positive, real, and no greater than unity. In the direction of boresight $G_{az,ant}(0) = 1$.

We identify the azimuth angle offset from the direction of the SRP for a flat target scene as

$$\phi_{az} \approx \text{asin} \left(\frac{|\hat{S}'_x|}{|\mathbf{r}_{c0}| + \hat{S}_{r,0}} \right). \quad (105)$$

In this case, we would define the correction to be

$$X_{correction,4}(u_y, u_x) = \frac{1}{G_{az,ant}(\phi_{az})}. \quad (106)$$

This correction is applied to each image pixel to yield the amplitude-corrected image

$$y'(u_y, u_x) = y(u_y, u_x) X_{correction,4}(u_y, u_x). \quad (107)$$

With this adjustment, the azimuth antenna pattern loss has been compensated.

While we have separated the antenna pattern loss corrections into elevation and azimuth corrections, a perfectionist might offer that the antenna beam pattern isn't strictly separable in this manner, and so the corrections are not separable either. Furthermore, we have presumed a flat target scene, which is also not generally strictly true, which will also impact the ideal correction.

While we stipulate that a more perfect correction of antenna beam characteristics is certainly possible, nevertheless the corrections described above are usually pretty good in that they typically serve to supply adequate radiometric calibration of the SAR images for most applications.

4.4.2 Image Spectrum Corrections

A number of SAR image exploitation schemes and post-processing algorithms (e.g. autofocus, etc.) require entering the spectral domain of the image. Sometimes this is in a single dimension, and sometimes this is in all dimensions. Consequently, the spectral characteristics of the image need to be understood, especially if they depart from what would otherwise be expected.

We begin this discussion with the observation that if we begin with a well-focused SAR image, then any phase function or perturbation that we apply to the pixel values does not alter the focusing of the image. The phase function or perturbation on the image pixels will, however, alter the spectrum of the image. Nevertheless, the image itself will remain focused.

So, the questions we have now are

1. What does the image spectrum look like?
2. What do we want the image spectrum to look like?
3. What do we have to do to take #1 and make #2 out of it?

4.4.2.1 What does the image spectrum look like?

Recall that we began with a data set that was presumed to model a spectrum that looks something like that in . We will assume that the residual frequency and chirp rates have been compensated. Note also that this spectrum is offset from DC. Recall that the model for this (ignoring the ramp filter) was given in Eq. (48) as

$$x'_V(i, n) = A_R \exp j \left(-\frac{2}{c} \gamma_0 T_s \kappa_n s_{r,n} i - \frac{2}{c} \omega_0 \kappa_n s_{r,n} + \frac{2\gamma_0 \kappa_n}{c^2} (s_{r,n})^2 \right). \quad (108)$$

Recall also the slant-range offset $s_{r,n}$ depends on coordinates s'_x and s'_y .

To understand the image spectrum, we need to know how the phase of $x'_V(i, n)$ varies with coordinates s'_x and s'_y . To examine this, we expand the phase in terms of some partial derivatives as follows

$$\Phi(i, n) \approx \left(-\frac{2}{c} \gamma_0 T_s \kappa_n i - \frac{2}{c} \omega_0 \kappa_n + \frac{4\gamma_0 \kappa_n}{c^2} s_{r,n} \right) \left[\left(\frac{\delta s_{r,n}}{\delta s'_x} \right) s'_x + \left(\frac{\delta s_{r,n}}{\delta s'_y} \right) s'_y \right]. \quad (109)$$

To gain some traction with this, we make the approximation $\kappa_n \approx 1$, which yields

$$\Phi(i, n) \approx \left(-\frac{2}{c} \gamma_0 T_s i - \frac{2}{c} \omega_0 + \frac{4\gamma_0}{c^2} s_{r,n} \right) \left[\left(\frac{\delta s_{r,n}}{\delta s'_x} \right) s'_x + \left(\frac{\delta s_{r,n}}{\delta s'_y} \right) s'_y \right]. \quad (110)$$

With respect to the center of the synthetic aperture, and the center of the resolution band, we calculate

$$\Phi(0, 0) \approx \left(-\frac{2}{c} \omega_0 + \frac{4\gamma_0}{c^2} s_{r,0} \right) \left[\left(\frac{\delta s_{r,0}}{\delta s'_x} \right) s'_x + \left(\frac{\delta s_{r,0}}{\delta s'_y} \right) s'_y \right]. \quad (111)$$

We may calculate the respective wavenumbers as follows.

$$\begin{aligned}
k'_y(0,0) &\approx \left(-\frac{2}{c}\omega_0 + \frac{4\gamma_0}{c^2}s_{r,0} \right) \left(\frac{\delta s_{r,0}}{\delta s'_y} \right), \text{ and} \\
k'_x(0,0) &\approx \left(-\frac{2}{c}\omega_0 + \frac{4\gamma_0}{c^2}s_{r,0} \right) \left(\frac{\delta s_{r,0}}{\delta s'_x} \right).
\end{aligned} \tag{112}$$

For our purposes, this may be approximated as

$$\begin{aligned}
k'_y(0,0) &\approx \left(-\frac{2}{c}\omega_0 + \frac{4\gamma_0}{c^2}s_{r,0} \right) \cos\psi_0, \text{ and} \\
k'_x(0,0) &\approx \left(-\frac{2}{c}\omega_0 + \frac{4\gamma_0}{c^2}s_{r,0} \right) \left(\frac{s'_x}{|\mathbf{r}_{c0}| + s_{r,0}} \right).
\end{aligned} \tag{113}$$

So, it is reasonable to ask “What does this tell us?” As an answer, we observe the following.

- The wavenumber (spatial frequency) component in the y' -direction is shifted by an amount proportional to ω_0 . This shift far exceeds the fast-time sampling frequency, so is generally observed as aliased to baseband.
- The wavenumber (spatial frequency) component in the y' -direction is shifted by an additional amount due to the RVPE. This additional shift is proportional to the range offset $s_{r,0}$. This range-offset dependent frequency shift is the principal effect of what we call ‘skew’.
- The wavenumber (spatial frequency) component in the x' -direction is ‘spread’ by an amount proportional to ω_0 , and proportional to s'_x .
- The wavenumber (spatial frequency) component in the x' -direction is spread by an additional amount due to the RVPE. This additional spread is proportional to the range offset $s_{r,0}$. This range-offset dependent frequency spread is a secondary effect of what we call ‘skew’.
- Essentially, the same data represents ‘different’ spatial spectral regions that depend on the particular pixel location in the image.

The significance of this is that during the course of preprocessing the data and image formation as described above, these various phase terms are compensated. This means that for the FBP algorithm as presented, the spectral regions are processed to be ‘aligned’ in the spatial frequency domain. This means that the image spectrum is shifted in both dimensions, meaning that the ‘aperture’ in wavenumber domain for one pixel in the image does not coincide with the ‘aperture’ in the wavenumber domain for another pixel

in the same image. This is illustrated in Figure 10. This effect is most noticeable for large images with respect to range. The key point is that ‘alignment’ is different than ‘coincidence.’ To be ‘aligned’ we have to mess up ‘coincidence.’ Figure 11 shows a synthetic image of four point targets, and their respective aligned contributions to the overall image spectrum.

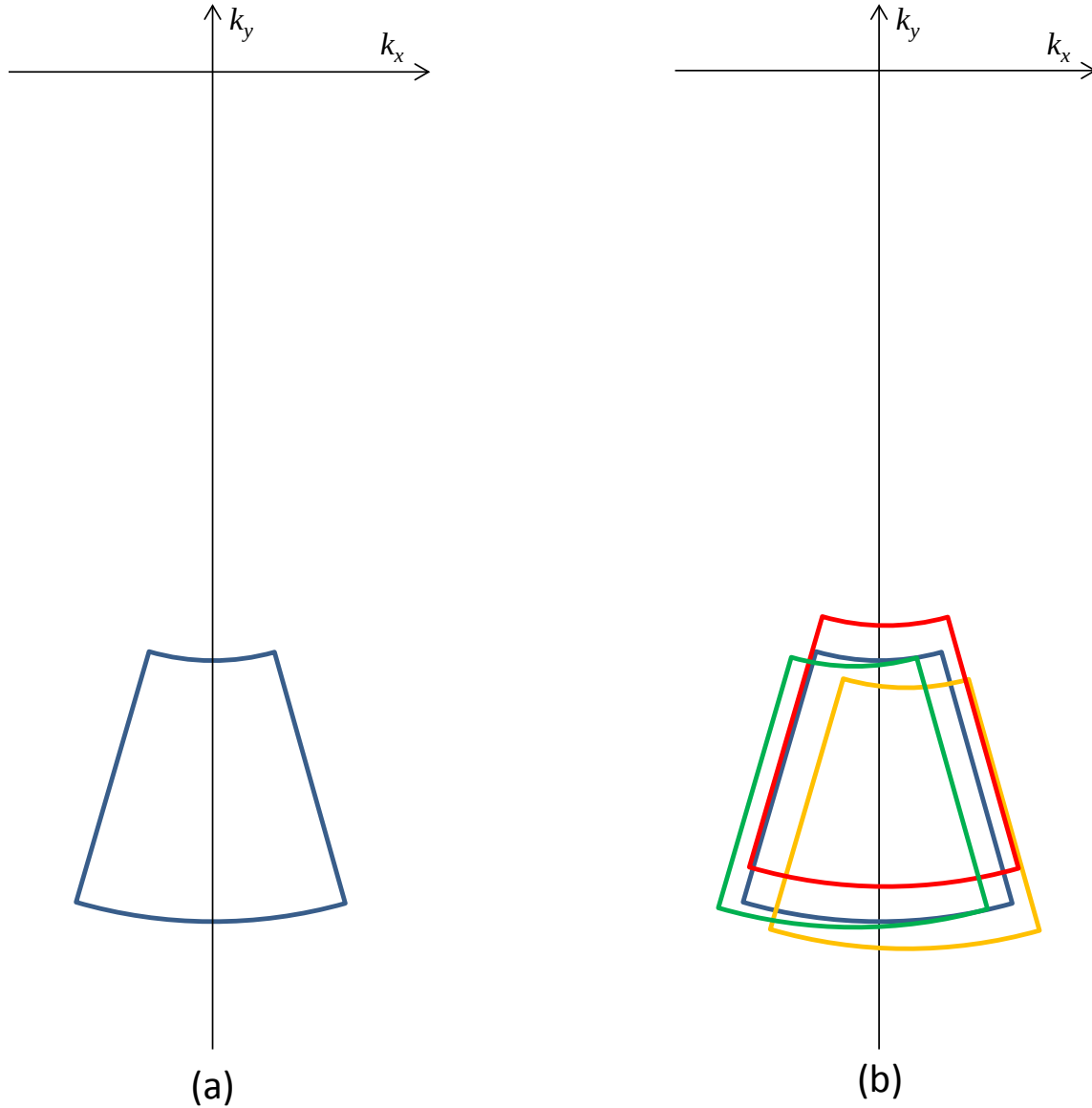


Figure 10. Notional illustration of (a) the often presumed image spectrum for a SAR image, and (b) the actual image spectrum for different pixel locations in a FBP processed SAR image.

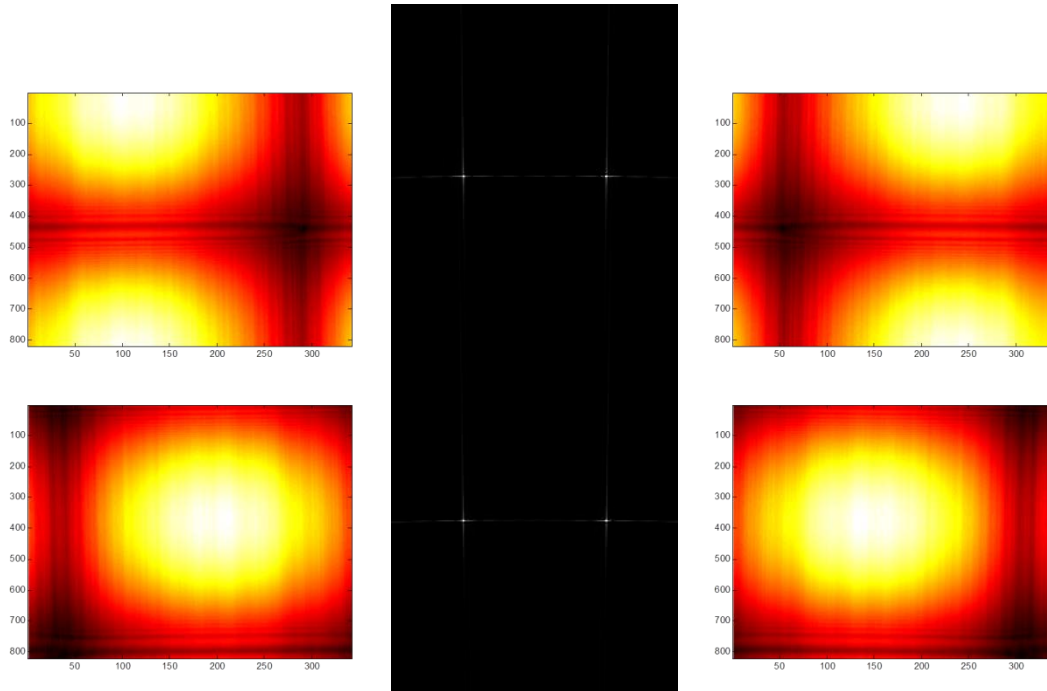


Figure 11. Synthetic image with four point targets, and their respective image spectra. Note that the spectra are shifted, dependent on target spatial location. The “blob” nature of the spectra is due to the window taper function used during image formation. DC of the raw phase-history data set is at the center of the spectra plots.

4.4.2.2 What do we want the image spectrum to look like?

Oddly enough, it is the completeness and fidelity of the matched filtering that the FBP algorithm implements that causes this effect to manifest as completely as it does. Other image formation algorithms often forego the ‘alignment’ of which we speak. These are sometimes referred to in the algorithms as ‘inconsequential’ phase errors, except that they aren’t entirely inconsequential. While foregoing the ‘alignment’ may keep the spectrums coincident, it also introduces phase and migration errors that limit the fidelity of the resulting image, and/or its spectrum.

Consequently, we may have a perfectly good image after FBP processing, but its spectrum may look substantially different than if having employed a different image formation algorithm.

This may be an issue for some post-processing and exploitation algorithms that are expecting coincident spectrums for each location in the SAR image. Consequently, we may at times wish for the SAR image to exhibit a spectral characteristic that does indeed look similar for all pixel locations, and perhaps even similar to the original data.

4.4.2.3 What do we do to make the pixel spectrums coincide?

We now present the operations to make coincident spectrums for each location in the SAR image. We will do this by ‘adding back’ some phase characteristics to the FBP formed image. Since the image is already formed, we have already benefitted from the focusing, and adjusting the phase of each pixel will have only the consequence of manipulating the image spectrum, and not the magnitude image itself.

Recall that the model for the original data (ignoring the ramp filter) was given as

$$x'_V(i, n) = A_R \exp j \left(-\frac{2}{c} \gamma_0 T_s \kappa_n s_{r,n} i - \frac{2}{c} \omega_0 \kappa_n s_{r,n} + \frac{2\gamma_0 \kappa_n}{c^2} s_{r,n}^2 \right). \quad (114)$$

The exact expression for phase is

$$\Phi(i, n) = -\frac{2}{c} \gamma_0 T_s \kappa_n s_{r,n} i - \frac{2}{c} \omega_0 \kappa_n s_{r,n} + \frac{2\gamma_0 \kappa_n}{c^2} s_{r,n}^2. \quad (115)$$

In an image, the best we can do is assume the indices are zero, which yields

$$\Phi(0, 0) \approx -\frac{2}{c} \omega_0 \kappa_0 s_{r,0} + \frac{2\gamma_0 \kappa_0}{c^2} s_{r,0}^2. \quad (116)$$

This then is the phase function that we wish to reinstate to the image. Consequently, our correction function becomes

$$X_{correction,5}(u_y, u_x) = \exp j \left(-\frac{2}{c} \omega_0 \kappa_0 \hat{s}_{r,0} + \frac{2\gamma_0 \kappa_0}{c^2} \hat{s}_{r,0}^2 \right). \quad (117)$$

This correction is applied to each image pixel to yield the spectrum-corrected image

$$y''(u_y, u_x) = y'(u_y, u_x) X_{correction,5}(u_y, u_x). \quad (118)$$

The spectrums of all image pixels should now be coincident, or at least very nearly so.

Figure 12 and Figure 13 show the results of this correction on the spectrum of a real SAR image.



Frequency = 16.7 GHz
Resolution = 3 m
Range = 15.37 km to SRP
Grazing Angle = 10.7 degrees
Squint Angle = 270 degrees

Figure 12. SAR image from Ku-band SAR data using FBP algorithm as detailed in this report. A coarse resolution large area image was chosen to exacerbate the image spectrum issues. The image is of an area near Lake Elsinore, CA. (Image courtesy of General Atomics, ASI.)

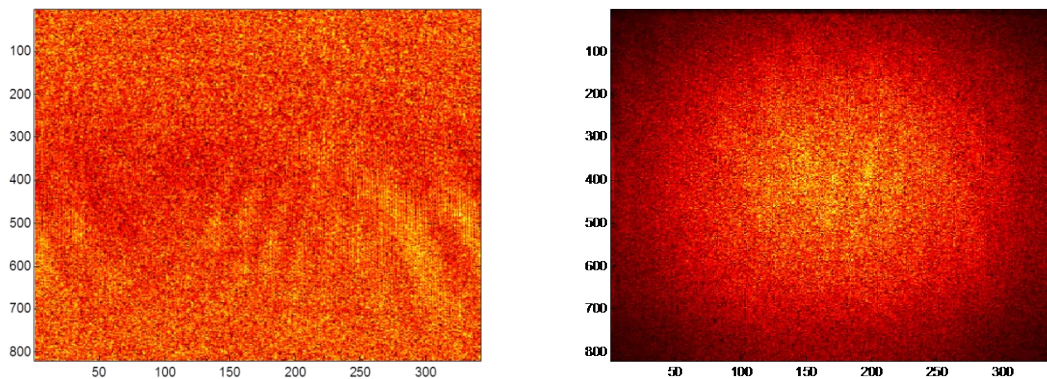


Figure 13. Image spectra from image of Figure 12.
(left) Aligned image spectrum. Image spectra are aligned, with different spatial locations in the image contributing to different spectral regions, with substantial aliasing exhibited.
(right) Coincident image Spectrum. All image spectra are coincident, forming a nice blob centered at DC. The taper towards the edges are due to the window taper functions used for sidelobe control.

4.4.3 Autofocus

In general, motion measurement in most SAR systems will not be adequate to facilitate maximally focused images. A report by Doerry discusses fundamental limits to the goodness of motion measurement systems.¹⁶ Furthermore, even a perfect motion measurement system cannot account for ‘electrical’ path length fluctuations due to atmospheric effects. Dickey, et al.,¹⁷ have shown that such fluctuations are quite common. This was confirmed by Lawrence, et al.¹⁸

This necessitates data-driven techniques for estimating and compensating the effects of unknown residual motion errors, including ‘apparent’ residual motion errors. Such techniques are termed “autofocus.” This is essentially a blind deconvolution of an unknown assumed phase error.

A variety of autofocus techniques exist and are thoroughly examined in the literature. Although a comprehensive treatment of autofocus algorithms is beyond the scope of this report, we do offer that one popular and effective technique is Phase-Gradient Autofocus (PGA).⁶ While PGA corrects phase errors, extensions can also correct residual migration errors.¹⁹

An underlying principal of most autofocus algorithms is that they treat ‘common’ errors in the image, or image region over which they are applied. If the spatial variance of the residual errors is too great, then the spatial variance needs to be addressed. One technique is to apply autofocus operations separately to different subimages (segments of the larger image). Another is to account for the deterministic aspect of the spatial variance in the application of the autofocus result.

Another almost universal assumption for autofocus is that the apparent motion error manifests principally as an azimuth (i.e. cross-range) phenomenon, which in most SAR images is aligned with a principal axis of the image. For an arbitrarily rotated image, where principal axes are not necessarily aligned with range and cross-range, then this becomes problematic. An elegant autofocus algorithm solution is not known to the authors at this time, and may indicate an opportunity for further research.

However, if image principal axes are indeed aligned with range and cross-range, then conventional autofocus techniques are expected to usually suffice.

“Self-education is, I firmly believe, the only kind of education there is.”
— *Isaac Asimov*

5 Conclusions

We reiterate the following points.

- FBP is a well-understood and well-documented algorithm for SAR image formation, for the most part.
- There are three principal processing steps (actually groupings of steps) for FBP processing. These are
 1. PH Data Preprocessing,
 2. Image Formation, and
 3. Image Post-Processing.
- Some individual processing operations may occur optionally in different ones of these groupings.
- There are processing techniques that can trade memory for processing operations in the data interpolation stage.
- The spectrum of an image created using FBP will exhibit alignment where spectral regions differ for different locations within the image. An error will need to be reintroduced to cause spectra to coincide, to facilitate techniques such as autofocus, etc.

“The more that you read, the more things you will know. The more that you learn, the more places you'll go.”

— Dr. Seuss, I Can Read With My Eyes Shut!

References

- ¹ Armin W. Doerry, "Computed Tomography – the details," Sandia Report SAND2007-4252, Unlimited Release, July 2007.
- ² D. C. Munson, Jr., J. D. O'Brien, W. Jenkins, "A tomographic formulation of spotlight-mode synthetic aperture radar," *Proceedings of the IEEE*, Vol. 71, No. 8, pp 917-925, August 1983.
- ³ LeRoy A. Gorham, Linda J. Moore, "SAR image formation toolbox for MATLAB", SPIE 2010 Defense, Security & Sensing Symposium, Algorithms for Synthetic Aperture Radar Imagery XVII, Vol. 7699, Orlando, FL, 5-9 April 2010.
- ⁴ J. L. Walker, "Range-Doppler Imaging of Rotating Objects," *IEEE Trans. on Aerospace and Electronic Systems*, Vol. AES-16, No. 1, pp 23-52, January 1980.
- ⁵ Armin W. Doerry, "Basics of Polar-Format Algorithm for Processing Synthetic Aperture Radar Images," Sandia Report SAND2012-3369, Unlimited Release, May 2012.
- ⁶ C. V. Jakowatz Jr., D. E. Wahl, P. H. Eichel, D. C. Ghiglia, P. A. Thompson, *Spotlight-Mode Synthetic Aperture Radar: A Signal Processing Approach*, ISBN 0-7923-9677-4, Kluwer Academic Publishers, 1996.
- ⁷ Walter G. Carrara, Ron S. Goodman, Ronald M. Majewski, *Spotlight Synthetic Aperture Radar, Signal Processing Algorithms*, ISBN 0-89006-728-7, Artech House Publishers, 1995.
- ⁸ William J. Caputi, Jr., "Stretch: A Time-Transformation Technique," *IEEE Transactions on Aerospace and Electronic Systems*, Vol. AES-7, No. 2, pp 269-278, March 1971.
- ⁹ Armin Doerry, "Bandwidth requirements for fine resolution squinted SAR," SPIE 2000 International Symposium on Aerospace/Defense Sensing, Simulation, and Controls, Radar Sensor Technology V, Vol. 4033, Orlando FL, 27 April 2000.
- ¹⁰ Edward Bishop, Robert Linnehan, Armin Doerry, "Video-SAR Using Higher Order Taylor Terms for Differential Range," Proceedings of the 2016 IEEE Radar Conference, Philadelphia, PA, USA, 02 - 06 May 2016.
- ¹¹ Robert N. McDonough, Anthony D. Whalen, *Detection of Signals in Noise – Second Edition*, ISBN 0-12-744852-7, Academic Press, 1995.
- ¹² Wilbur B. Davenport, Jr., William L. Root, *An Introduction to the Theory of Random Signals and Noise*, ISBN 0-87942-235-1, IEEE Press, 1987.
- ¹³ Armin W. Doerry, "Performance Limits for Synthetic Aperture Radar – second edition," Sandia Report SAND2006-0821, Unlimited Release, February 2006.
- ¹⁴ A. W. Doerry, E. Bishop, J. Miller, V. Horndt, D. Small, "Designing interpolation kernels for SAR data resampling," SPIE 2012 Defense, Security & Sensing Symposium, Radar Sensor Technology XVI, Vol. 8361, Baltimore MD, 23-27 April 2012.
- ¹⁵ Armin W. Doerry, "Anatomy of a SAR Impulse Response," Sandia report SAND2007-5042, Unlimited Release, August 2007.
- ¹⁶ Armin W. Doerry, "Motion Measurement for Synthetic Aperture Radar," Sandia National Laboratories Report SAND2015-20818, Unlimited Release, January 2015.
- ¹⁷ Fred M. Dickey, Armin W. Doerry, Louis A. Romero, "Degrading effects of the lower atmosphere on long range airborne SAR imaging," *IET Proceedings on Radar, Sonar & Navigation*, Vol. 1, No. 5, pp. 329-339, October 2007.
- ¹⁸ Michael E. Lawrence, Charles T. Hansen, Samarendra P. Deshmukh, Brent C. Flickinger, "Characterization of the effects of atmospheric lensing in SAR images," SPIE 2009 Defense & Security Symposium, Radar Sensor Technology XIII, Vol. 7308, p 73080C, Orlando FL, April 29, 2009.
- ¹⁹ Armin W. Doerry, "Autofocus Correction of Excessive Migration in Synthetic Aperture Radar Images," Sandia Report SAND2004-4770, September 2004.

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